

OSHMITA SARKAR

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EDUCATION

KIIT UNIVERSITY

Bachelor of Technology in Computer Science and Engineering, Honor

Sep 2020 – Apr 2024

CGPA: 8.84/10

ARMY PUBLIC SCHOOL, BKP

Higher Secondary (AISSCE), PCMB

Mar 2020

95.60%

EXPERIENCE

AMAZON WEB SERVICES

May 2025–Present

Software Dev Engineer

Berlin, Germany

- Drove AWS Management Console's dual-stack migration across ~**1,432** domains, reducing cross-console regression rates by **62%** across regions and services. Strengthened the testing infra with E2E tests that caught **45+** regressions.
- Re-architected Lambda error-stack deobfuscation to solve a Linux-specific memory bottleneck, combining algorithmic optimization with an isolated, self-healing Piscina worker pool to reduce memory utilization from **120% to 60%** in major AWS regions and latency by **15%**. Won the Amazon OpEx 2026 award for saving **48** hours diagnosis efforts.
- Led the launch of company-wide domains and fallback infrastructure across **6** isolated U.S. regions, coordinating **3** AWS teams to extend platform capabilities to previously unsupported environments.
- Fast-tracked for progression to **mid-level engineer in 15 months** for sustained cross-team technical impact. Regularly supported **6+** teammates through technical mentorship and knowledge sharing.

FORMA.AI

Jun 2024–Mar 2025

Analytics Engineer

Pune, India

- Engineered Spark and AWS-based big-data solutions serving **10,000+ multi-region users and partners**. Resolved a **20%** commission-runtime breach by migrating report generation to Databricks, reducing execution time by **6** hours (~**30%**) and preventing failures in the **20-hour** workflow.
- Optimized data and application performance through SQL query redesign, modular pipeline architecture, and schema optimization, reducing query runtime from **20s to 0.37s**, pipeline runtime from **24s to 11s**, and data processing runtime by **6 hours**. Created ~**10** modules and onboarded new joiners to SQL, Git, Docker, and mentored existing teammates.
- Automated processing of **10+ monthly base-plan templates** using Redis, Hive Metastore, and Databricks SQL, improving data-processing efficiency by **25%** and enabling scalable analytics across distributed teams.

PUBLICATIONS

QSAR FOR DRUG DISCOVERY TARGETING THE GLUCAGON RECEPTOR USING MACHINE

LEARNING | 10th IEEE UPCON 2023, Amity University

[\(Publication\)](#) [\(Certificate Link\)](#)

- Built a Random Forest model on ChEMBL data for glucagon receptor ligand-affinity prediction, achieving an R² of **0.361**. Conducted statistical modelling experiments on the ChEMBL dataset for 7 days.

ANPR USING PADDLE-OCR | Presented in ICICET 2024

[\(Publication\)](#) [\(Certificate Link\)](#)

- Pre-processed **1000+** images, trained our custom Resnet50 neural network and tuning Paddle-OCR, achieving an average of **91.06%** confidence percentage; hypothesis testing signifying (p-value < **0.05**) over baseline models.

PROJECTS

MEDI-ASSIST | Streamlit, Python, Docker, Open AI API

[\(GITHUB LINK\)](#) [\(VIDEO DEMO\)](#)

- Developed a Streamlit-based medical assistant integrating OpenAI APIs to simplify medical terminology for patients and doctors.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, SQL, R

Cloud & Data: AWS, Spark, Databricks, PostgreSQL, Redis, PySpark

Systems & Engineering: Distributed Systems, Data Engineering, Docker, Git, CI/CD

ACHIEVEMENTS

- **Top 8 finalist**, WABTEC EXCEED 2.0 Challenge, among **1,500+ participants**; Project: Machinery hazard detection.
- **1st Runner-Up**, Inspirathon Hackathon, NSIT Delhi. Project: Built an anti-cyberbullying chatbot on discord.
- **Moderator**, G20 India Hackathon 2023, overseeing **200+ participants**. LOR from the director. More on hackathon.